

Behavior of Atmospheric Air Pollutants during Diwali Fireworks Emission over Western Tropical Semi-Arid and Sub-Urban site, Udaipur, India

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Case Report

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Abstract

Influence of fireworks emissions activity during “The Festival of Lights” or Diwali Episode on the simultaneously measured atmospheric trace gases including suspended materials as air pollutants, i.e., Surface ozone (O_3), Nitrogen Oxides (NO_x), Carbon monoxide (CO), Particulate Matters of size less than $2.5\mu m$ ($PM_{2.5}$) along with derived sources identification parameters concerning to atmospheric pollution, i.e., angstrom absorption coefficient of Black Carbon aerosol (a_{abs}) and CO/ NO_x values have been investigated during Diwali and prior as well as in post-Diwali days over Udaipur. From induced ambient air pollution effect during Diwali episode on air pollutant strength and the source discrimination parameters, the following main interesting facts are obtained: (i) a significant amount of enhancement by two to three-time folds in $PM_{2.5}$ level (peak up to $165 \mu g m^{-3}$) noticed in Diwali festival day relative to background condition level, i.e., greater than the double value of National Ambient Air Quality Standards (NAAQS) of $PM_{2.5}$, (ii) specifically, during after sunset to pre-midnight hours, air pollutants along with their source identification parameters values revealed an enhancement by ~ 1.5 to 3 times more during the festival days relative to their non-festival levels resulting from the impact of extra intense spontaneous fireworks and sparklers emission activities in a coincidence of their period with the peak festival celebration. However, their highest magnitude did not go beyond their healthy prescribed limit values, (iii) only temporal day hour O_3 variations showed almost appreciable enhancement (varied from 10 to 68 ppb) in conjunction with a remarkable reduction in NO_x (spanned from 20 to 2 ppb) in Diwali period and (v) about 40% enhancement in source discrimination parameters values displayed a clear demonstration of majority source type as bio-mass burning activity in Diwali episode and fossil fuel combustion activity in non-festival days. Results are also explained basis on bringing air pollutants by backward airmass trajectory starting from IGP source to receptor site as regional atmospheric air pollution effect.

1. Introduction

In recent years, a considerable attention has been given on the local air pollutant monitoring as concerned to the short-term elevation in atmospheric pollutants level. Since air pollutants are released into the earth's atmosphere, specifically during the bio-mass burning and fossil-fuels combustion activities concerning to both natural as well as human-made activities (Senifeld and Pandais (2012); Jethva et al. (2019); Ganguly et al. (2019); Kumar et al. (2016); Veechi et al. (2008); Wang et al. (2007); Mandal et al. (1997, 2012); more references there in). Some of such typical intensive burning events are of great concern to air pollution degradation (i) wild grass and crop residues burning over agricultural land regime (ii) incomplete combustion of fossil fuel in automobiles, industrial activities etc., over surrounding regions (iii) fire ignition crackers and sparklers, incense burning in temples and lighting activity during the various festival celebrations period in adjacent areas.

Among all of these, a shorter time scale perturbation response in air quality degradation level in Diwali festival celebration period has been widely recognized over last decade as one of the unique anthropogenic events, which has been regularly experienced in either October or November month of every year over the Indian subcontinent.

Numerous earlier researchers from their investigations have established the fact of the substantial amount of enhancements in ambient atmospheric toxic and health hazard air pollutant levels (O_3 , NO_x , CO, SO_2 , BC aerosol mass concentration, and $PM_{2.5}$, PM_{10} along with the release of harmful toxic chemical mixtures, etc.), by varying their magnitudes from about 2 to 6 time folds or more as well as an unusual rise in observed and derived microphysical optical properties of total atmospheric columnar coarse and fine size aerosols parameters (Aerosol Optical Depth (AOD), Single Scattering Albedo (SSA), etc., by varying in their ranges between 20 to 40% from their normal day level than their detected levels in several firework occasions, such as the Diwali festival, Lass Falls in Spain, Lantern festivals in Beijing, New-Year celebration and bio-mass burning activity of agricultural crop-residue materials etc., all over the globe as well as around several parts of the Indian sites too (Ravindra et al. (2003); Kulshrestha et al. (2004); Veechi et al. (2008); Badrinath et al. (2009a, b); Mandal et al. (2012); Vyas and Saraswat (2012); Vyas et al. (2012b); Nishant et al. (2012); Yerramsetti et al. (2013); Pathak et al. (2013); Pachauri et al. (2013); Saha et al. (2014); Devara et al. (2015); Mukherjee et al. (2018); Shivani et al. (2019); Ressmi et al. (2019); Chhabra et al. (2020)).

Moreover, their individual investigations on significant enrichment in air pollutants level also highlighted the impact on the reduction of net incoming solar radiation through scattering and absorption mechanisms with various types of ambient air pollutants and aerosols as their direct effect (Singh et al. (2010); Singh et al. (2014); Chhabra et al. (2020)). However, its indirect influence is visualized in the way of the alteration of the cloud condensation nuclei properties through absorption of incident sun's radiation in a broad spectrum from visible to infrared wavelength range due to extra emission of BC aerosols and trace gases in Diwali celebration resulting in warming to the cloud in the atmosphere or at the top of the atmosphere, i.e., known as cloud-aerosol interaction and leads to formation of dense cloud. This may become as one of major causes of lowering the atmospheric visibility (Ravindra et al. (2003); Yerramsetti et al. (2013); Saha et al. (2014); Pratap et al. (2021)).

It, furthermore, triggers to smoke, mist and fog formation activity under suitable prevailed meteorological conditions coupled with atmospheric boundary layer dynamics processes and their atmospheric secondary processes. As a result of this, it has become a wider concern of direct visible effect by reducing the atmospheric visibility level below less than one km from the prescribed healthy limit of standard values of 10km and, hence, also contributed as one of the severe visible effects on the earth's surface owing to air pollution degradation level. Because of these, they are also capable to perturb the meteorological conditions for a few hours to days period in such evens and after the events or vice-versa (Jayaratne and Verma, 2001; Saha et al., 2014; more references therein).

Thus, despite their above great current relevance in deteriorating air quality level and public human and plant life as well as atmospheric visibility, etc., very limited specifically simultaneous measurements of the surface O_3 , NO_x , CO, BC, $PM_{2.5}$ levels during the eve of Diwali celebration for their source strength variability have been conducted over sub-urban tropical cities of lesser populated western Indian location. Moreover, the analysis of identification of an additional released of air pollutants by various kinds of sources either by bio-mass burning or due to fossil-fuel combustion activities during the Diwali festival has been also not touched much so far in the previously reported investigations, and, hence such types of studies are either sparse or rare over the tropics so far in India.

Therefore, owing to its serious short term air pollution degradation issue relevant to firework displays celebration and its consequent impact on earth-atmosphere solar radiation, human health hazard, atmospheric visibility, etc., the influence of firework activities on simultaneously measured atmospheric air pollutant concentrations, such as O_3 , NO_x , CO, BC, and 24 hourly mean $PM_{2.5}$ over Udaipur (24.6°N, 73.6°E, ALT. 580m, above sea level (ASL)) have been discussed in this paper for pre-Diwali onward to main Diwali and post-Diwali festival period, i.e., from 26th Oct. to 13th Nov. 2010.

It is to be noted that derived source type parameters such as, α_{abs} and CO/ NO_x have been extensively adapted as source identification variables of air pollutants related to either contributing from fossil fuel incomplete combustion and biomass burning material activities. The (i) α_{abs} parameter which is computed from the negative slope of the linear regression line of the log of BC absorption coefficient value (BC_{abs} , Mm^{-1}) at seven wavelengths and log of its wavelength (μm), and (ii) ratio from CO to NO_x concentration (Kaskaoutis et al. (2011); Anjeja et al. (2001), more references there in).

At the same time, the altering in the nature of the shape or slope of the spectral wavelength dependence BC_{abs} curve, i.e., α_{abs} is served as a crucial derived indicator of distinguishing the relative contribution of anthropogenic BC aerosols generated from the mainly two kind of sources, first by ejecting from incomplete combustion from fossil fuel materials source activities i.e., automobile exhausts as diesel soot particles, etc., where α_{abs} 's values < 1 and second by production of air pollutants by burning of bio mass materials, where, α_{abs} 's values > 1 (Bergstrom et al. (2002); Kirchstetter et al. (2004); Kaskaoutis et al. (2011)). Likewise, the escalating of CO/ NO_x values gave the furthermore inferences about relative change in potential source's types of air pollutants released owing to more predominance biomass burning over the fossil fuel combustion emission activity.

Therefore, in view of the identification of the possible emissions of air pollution sources and distinguished their role in the one of major atmospheric pollution event compared to normal period has been aimed in the present study by elaborating the interrelation between change in the temporal evolution of source identification parameters, i.e., α_{abs} , and the ratio of CO to NO_x from preceding Diwali to the proceeding Diwali periods.

As such types of this study about segregating the contribution of the relative shares of varieties source emission characteristics along with air pollutants strength during the festival celebrations would give further more insights to enhance our scientific understanding and its conformation about major role of local and regional air pollutants emission source types and their relative strength variability in such specific anthropogenic air pollutant event over sub-urban and less air populated region.

2. Materials And Methods

2.1 Experimental Setup and Observation Schedule Details:

The synchronized surface observations of surface O_3 , NO_x (NO , NO_2), CO, $PM_{2.5}$, and BC aerosols mass concentrations were started at the Department of Physics, Sukhadia University, Udaipur under DST and ARFI (ISRO-GBP) research project since December 2009. Simultaneous measurements of above atmospheric ambient pollutants were carried out regularly and round the clock of 24 hourly observations using precise atmospheric trace gas analyzers, such as, Ozone analyzer (Model 49-I, Thermo Scientific, USA), NO_x analyzer (Model 42-I TL, Thermo Scientific, USA), CO analyzer (Model 48-I TLE, Thermo Scientific, USA), Fine particulate sampler (Model APM-550, M/s Envirotech Ins, Delhi) and Seven Channel Aethalometer (Model AE-31, Magee Scientific Company, USA). The lowest catching limits of these trace gases analyzers are 1 ppb (part per billion) for O_3 , 0.4 ppb for NO_x (linearity 1%) and 50 ppb with linearity 1% for CO to conduct the present work.

Since all such measuring equipments are world-wide accepted as standard, popular, and reliable readymade and fabricated instruments by several reputed commercial manufacturers. At the same time, their principles, techniques and technical specifications of detecting the trace air inorganic pollutant level by these instruments are also as per recommendations based on technical specifications prescribed by the USEPA (United State Environmental Protection Agency; Demerjian, (2004)) guidelines. To maintain the similarities, all the equipment employed for the measurement of a secondary pollutant O_3 and the primary pollutants NO_x , and CO at the sampling site has been employed of the same company.

Moreover, most of the similar kinds of earlier studies have also been performed by employing such identical types of trace gas analyzer worked on the same principle and technical specifications over various Indian sites by several investigators (Renuka et al. (2014); Yerramsetti et al. (2013); Nishanth et al. (2012); Beig et al. (2007); more references are therein).

The 24-hourly intervals of $PM_{2.5}$ samples were collected on 47 mm pre-weighted PTFE (Poly Tetra Fluro Ethylene) filters using the standard instrument High volume sampler. Based on the Gravimetric method, the mass concentration of $PM_{2.5}$ in the air sample is computed as the total mass of collected particles divided by the volume of air sampled and is expressed in micrograms per cubic meter of air. The lower detection limit of the mass concentration measurement range is approximately $2 \mu g m^{-3}$ as per reported by the manufacturer.

The continuous observations of surface mass concentration BC aerosols from UV to IR range were taken by the well-known popular, standard reliable, and near-real-time instrument, i.e., Aethalometer (relative errors about 10% as reported by the manufacturer). Furthermore, technical details of the instruments with their models, companies along with their calibration details, etc., utilized in this study has been already discussed elsewhere and also summarized in Table 1 (Vyas et al. (2012a); Gaur et al. (2014)).

Table 1

Details of instruments employed for measurement of various ambient air pollutant concentrations, etc., at the monitoring site:

S.No	Name of Parameters with their units	Instruments used with their manufacturer's name	Model
1.	Surface Ozone (O_3 , ppb)	Ozone analyzer, (M/s Thermo Scientific, USA)	Model 49-I
2.	Nitrogen Oxides (NO_x , NO_2 , NO , ppb)	NO_x analyzer (M/s Thermo Scientific, USA)	Model 42-I TL
3.	Carbon monoxide (CO , ppb)	CO analyzer (M/s Thermo Scientific, USA)	Model 48-I TLE,
4.	Particulate Matters below 2.5-micron meter ($PM_{2.5}$, $\mu g\ m^{-3}$)	Fine Particulate Sampler (M/s Envirotech, Ltd., New Delhi)	Model APM-550
5.	BC aerosols mass concentration (BC , $\mu g\ m^{-3}$)	Aethalometer Seven Channel (M/s Magee Sci.CA., USA)	Model AE-31
6.	The three hourly interval data of meteorological and boundary-layer parameters such as T, Rh, WS, WD, PBL and 72 hourly air mass backward wind trajectory at 500m and 4000m above the ground level.	Downloaded from the NASA website, i.e., http://www.ready.noaa.gov .	

Regular and systematic observations were originally planned to record the air pollution concentration levels, except the $PM_{2.5}$, originally in an interval of five minutes during the period from October 26 to November 13, 2010. The preceding five days interval from 26th Oct to 31st Oct. 2010 were selected as before Diwali days and the succeeding five days period from 7th Nov., to 13th Nov. 2010 was considered as after Diwali day period. However, consecutive days from 1st to 6th Nov 2010, it was treated as Diwali days. As, Diwali festival activities are generally celebrated not only for one day, but their celebrations of fire ignition event period can be extended from three to six days.

Due to the non-availability of scientific instruments for measurement of local general meteorological parameters as secondary data, three hourly resolution interval values of each day time series of such meteorological and boundary-layer parameters such as surface temperature (T , $^{\circ}C$), relative humidity (Rh , %), wind speed (WS , ms^{-1}), wind direction (WD , measured in degree clockwise direction from North) and mixed boundary layer height (PBL , m) of preceding, succeeding, and during Diwali days over similar geographical position within a grid of 0.5×0.5 -degree level was accessed from NOAA READY (National Oceanic and Atmospheric Administration Real-Time Environmental Applications and Display System) archived GDAS (Global Data Assimilation System) server, i.e., <http://www.ready.noaa.gov>, which are employed during the course of this investigation as similarly used by earlier researchers also in their studies too at different Indian locations (Soni et al. (2019); Singh et al. (2015); Tiwari et al. (2013); Vyas et al. (2021)).

2.2 Study Area and Prevailing Meteorology:

2.2.1. Study area description:

Figures 1(a & b) depict the geographical location of the sampling site, i.e., Udaipur city, which is situated in a western Indian region of the southern Rajasthan and adjacent to Gujarat and Madhya Pradesh state border of India. Udaipur is a semi-arid (total overall annual rainfall ~ 580 mm) and a typical suburban with less air polluted tropical city of having a total district population of about three million as well as total city population around 0.5 million only, according to the latest census reports (<http://www.census2011.co.in>).

As per the census report of 2011, the majority populated area about 80.18% belongs to rural population types and the remaining populated region, i.e., 19.82% concerns to urban activities. The geographical region is positioned near the Great Indian Thar Desert toward NW from site, and it also lies in the foothills of the southern Aravali Mountain region. It has an average elevation of 580 m, ASL.

The monitoring of several air pollutants was carried out on the terrace (about 15 m above the ground) of the First Floor of Physics Department building in University College of Science Campus, Udaipur, which is surrounded nearby the prime location of densely residential colonies and commercial industrial marketing and industrial activity area of Udaipur old city (as shown in Fig. 1c). Thus, it has concerned of many humans, traffic, and vehicular and industrial emissions as anthropogenic activities regimes within the circumference of about 10 km diameter. The total number of different types registered automobile vehicles in Udaipur district is around 0.68 million, which comprises 76% of two-wheelers, 8.6% of cars, 3.4% trucks and the remaining 12% of other types of vehicles, such as tractors, buses, jeeps, tempos, etc. Thus, the main emission sources of ambient air pollutants primarily belong to domestics, vehicular and industrial activity sectors, which comprises of bio-mass, and fossil fuel emission activities sources. Further details of the site and air pollutants sources have been discussed in earlier works (Hegde et al., (2020); Vyas et al. (2021)).

To evaluate the influence on atmospheric ambient air quality level induced due to firecrackers and sparkers burning activity during the Diwali period, special systematic Diwali days variation of O_3 , CO , NO_x , BC and $PM_{2.5}$ levels along with meteorological and atmospheric boundary layer parameters have been inter compared to quantify the role of firework induced influenced on the similar temporal evolution of the source discrimination analysis parameters, such as ratio CO to NO_x and α_{abs} , during the above episode period.

2.2.2. Description of Prevailing Local Meteorological Condition:

Considering their potent role of prevailing meteorological processes and boundary layer dynamics activity, such as atmospheric solar heating, wind dispersion, accumulation, transportation phenomena on air pollutants level, the overall average temporal hourly behavior in day and night time of the above four concerned local metrological parameters, such as T, Rh, VC (Ventilation coefficient; the product of wind speed and planetary boundary layer height; m²/sec) and WD, are plotted from time-series data for continuous 18 days in same Figs. 2(a-d) and Figs. 3(a-d), respectively, and their computed statistical variable values are presented in Table 2.

Table 2

Statistical parameters of related meteorological parameters computed during Diwali, after and before Diwali Festival Day and Night hours period.

Day	Monitoring Dates	T (°C)	T (°C)	Rh (%)	Rh (%)	WD Day Hours	WD Night hours	VC (m ² /Sec)	VC (m ² /Sec)
		Day Hours	Night Hours	Day Hours	Night Hours	° from North)	° from North)	Day Hours	Night Hours
		Mean ± SD	Mean ± SD	Mean ± SD	Mean ± SD			Mean ± SD	Mean ± SD
		(Min, Max)	(Min, Max)	(Min, Max)	(Min, Max)			(Min, Max)	(Min, Max)
Before Diwali	27 Oct to 1 Nov, 2010	29.6 ± 3.7 (23.1,35.1)	21.4 ± 2 (17.2,24.9)	24.5 ± 6.6 (13.8,36)	37.8 ± 6.1 (26.3,49.2)	94 ± 93.7 (6.2,329.1)	48.6 ± 19.5 (13.8,86.2)	4701.8 ± 4712.5 (314.9,15054.)	339.6 ± 178.7 (57.8,625.9)
Diwali	2 to 7 Nov,2010	28 ± 3.8 (21.8,33.3)	19.2 ± 2.1 (16,23.3)	24.2 ± 7.7 (14.6,41.3)	39.7 ± 6.7 (29.4,53.8)	70.6 ± 21.6 (39.3,134.8)	68.3 ± 22.9 (31.1,130)	4583.3 ± 4308.2 (221.5,12192.)	359 ± 297 (12.2,1101.7)
After Diwali	8 to 13 Nov,2010	28.9 ± 2.9 (22.1,33)	23.5 ± 1.5 (21.1,26.1)	49.3 ± 9 (37.4,65.1)	65.9 ± 8 (42.8,81.1)	133.2 ± 81.3 (14.3,284.9)	118.6 ± 100.8 (0,295.8)	2925.8 ± 2970.4 (90.8,12230.9)	273.5 ± 335.6 (14.7,1345.1)

(i) Fig. 2a and 3a showed the systematic and regular marginal reduction trend in both ambient maximum temperature in midday hours around 11:00 hr and minimum hours temperature values around 08:30 hr by about 2°C only, i.e., from 35.1°C to 33.3°C and from 23.1°C to 21.8°C, respectively, relative to normal to festival days. During the night hours period, their maximum and minimum temperature values were also found to be declined in a higher magnitude of 3°C to 5°C, i.e., in the range from 23.3°C to 26.1°C and 16°C to 21.1°C, over similar period, respectively. Hence, during the Diwali festival period, the day and night hours surface temperature fluctuations showed a slight reduction of 2 to 5°C, respectively, implying clear evidence of the colder environment in nocturnal period relative to sunlight hours over festival relative to normal period.

(ii) From a close look to Fig. 2b and 3b, the highest Rh levels in the morning around 08:30 hr were noticed to be consistently increased by almost double magnitudes from 35.8–65% over before-Diwali to after-Diwali period and enhancement of three times fold of Rh value from 14.6 to 41.3% in the main Diwali days.

While in the nocturnal duration, peak Rh values transmuted marginally by the order of 4% only in between varying from 49.2–53.8% during pre to main Diwali period. However, an unusual enhancement corresponding Rh values shot up to a peak value of 81% or more than five times, which was visible during preceding festival eve. In contrary to this, the lower mean Rh values in Diwali and before Diwali days (31 ± 1.8% and 30.5 ± 8.8%) were also seen to be lesser humid period by about 28% than yielded higher Rh value of 58.3 ± 12.3% after the Diwali event. Thus, from the ongoing analysis and from Figs. 2b, 3b and Table 2, it is evident that a lesser humid environment is observed in the before to Diwali episode period relative to more humid conditions in past-Diwali occasion.

(iii) As VC is a good combined meteorological and atmospheric planetary layer dynamical variable and indicator of giving information about spreading and accumulating power efficiency of air pollutants takes place with varieties of several atmospheric meteorological dynamic processes only itself, i.e., atmospheric transport, convection, and dispersion phenomena, etc. (Nair et al. (2007); Sujata et al. (2016)). It is quite obvious from the temporal variation of the day and night hours of VC as depicted in Fig. 2(c) and Fig. 3(c) that, in the case of daytime, the lowest VC values have been systematically declined from their peak value 314.9 m²/s in pre-Diwali, followed by their intermediate value of 221 m²/s and their trough value of 90.8 m²/s in post episode days.

While during night hours, the order of descending trend in the minimum VC values, which are lesser as compared to day hours as follows: highest (57.8m²/s) in pre-Diwali >(14.7m²/s) in post-Diwali days, and the lowest magnitude (12.2m²/s) only after Diwali events. Based on the comparative analysis of day and night period observations in the celebration period, night time values always remain in the lower magnitude representing the more atmospheric instability condition or lesser dispersive power, i.e., more accumulation efficiency of air pollutants attributed to local meteorological and boundary layer dynamics activities in the absence of solar radiation than during the sunlight hour's conditions.

(iv) Figs. 2 (d) and 3 (d) illustrate continuous time series of surface wind direction of day as well as night hours for pre-to post-Diwali through the Diwali event period. It is seen from the plot as well as from table that during the day and night hours of the pre-Diwali event, surface wind directions remained predominantly north and northeasterly, respectively. While in Diwali period, their preferred wind direction was northeasterly and easterly. While after Diwali event, it was observed of northerly to north-westerly.

Thus, most probable ground wind directions flow pattern is from the NE and E sectors throughout the day and night hours during the Diwali event, but, it is appeared to be confined and turned towards the N and NE regions in pre-Diwali and bowed from N and N and NW sites in post-Diwali.

In concise from the relative change observed in the above meteorological parameters, etc., during festival and non-festival occasion, the reduction in ground-level temperature changes only by 2 to 5°C during the Diwali day and night hours, whereas, the maximum Rh levels are found to significantly reduced by 27% in night hours and only 5% in day hours from Diwali to after-Diwali. Whereas in the case of minimum VC levels, the values are always seen to be lower in the night hours in comparison with their day hours values. Furthermore, overall day and nighttime wind direction patterns are predominantly northeasterly in Diwali, southeasterly in post-Diwali, while in pre-Diwali, the mean wind becomes easterly in the day hours and turned in to northeasterly during the night hours.

3. Results And Discussion

3.1. Fine size suspended air particulate matters PM_{2.5} mass concentration level:

The primary sources of ejecting atmospheric air pollutants are being considered due to the ignition of the firework activity of biomass and fossil fuel materials in the Diwali episode. Therefore, mixed gasses along with material pollutant levels are expected to be enhanced significantly as the result of such extensive burning of firecrackers and sparklers and lighting display activities on the eve of Diwali festival.

Therefore, in order to assess the impact on the air suspended particulate substance level on such special anthropogenic occasion, the day-to-day average PM_{2.5} variations have been plotted with respect to pre-Diwali, on Diwali, and post-Diwali days in Fig. 4. It is seen from the plot that day to day PM_{2.5} concentrations vary gradually rises from 34.9 μgm^{-3} to 95.35 μgm^{-3} (day to day variability of 180%) during before Diwali days, i.e., from 27 Oct to 1 Nov 2010, demonstrating the reasonable influence of starting the firework, sparklers and crackers celebration activity as well as biomass materials burning activity on the PM_{2.5} level on the pre-festival period.

Subsequently, onward main Diwali days of peak anthropogenic emission activities concerning to widespread use of bursting the fire-crackers, sparklers, light decorations, etc., as the result of this, daily PM_{2.5} concentration levels were noticed to be enhanced abruptly ranging from 85 μgm^{-3} on 2^{ed} Nov and to be attained their abnormal peak value of 168 μgm^{-3} on the main Diwali. Later, the day's average PM_{2.5} level illustrated their sudden reduction from 109.4 μgm^{-3} to 45.6 μgm^{-3} on 6th to 8th Nov with the concurrence of lesser activities of burning of sparklers and other ignition activities just after Diwali. Hence, such a sharp increase in 24 hourly average PM_{2.5} levels by about 3 times was registered in the prime Diwali festival day relative to control days. However, after the Diwali festival, their values again continuously decline and retain their normal daily values from 31.1 μgm^{-3} to 54.6 μgm^{-3} with the coincidence of the rare occurrence of firework celebration activities after the festival.

Hence, from the above discussion and computed statistical parameters, i.e., mean with standard deviation, maximum and minimum values as shown in Table 3, it is quite imperative to say that the comparative change in the mean fine particulate matters of PM_{2.5} level has shown an abnormal rise by greater than 2.5 times above from the prescribed safer limit of NAAQS's (PM_{2.5} magnitude of 60 μgm^{-3}) and well exceeding of 5 time folds from the USEPA (35.0 μgm^{-3}) standards on Diwali.

Table 3
Average Day and Night hours values with their standard deviations and their minimum and maximum observed air pollutants values, i.e., O₃, NO_x, CO, CO/NO_x a_{abs} during before Diwali, Diwali and after Diwali festival event period.

Period	Monitoring Dates	Ozone (ppb) Day Hours Mean ± SD (Min, Max)	Ozone (ppb) Night Hours Mean ± SD (Min, Max)	NOX (ppb) Day Hours Mean ± SD (Min, Max)	NOX (ppb) Night Hours Mean ± SD (Min, Max)	CO (ppb) Day Hours Mean ± SD (Min, Max)	CO (ppb) Night Hours Mean ± SD (Min, Max)	CO/NOx Day Hours Mean ± SD (Min, Max)	CO/NOx Night Hours Mean ± SD (Min, Max)	a _{abs} Day Hours Mean ± SD (Min, Max)	a _{abs} Night Hours Mean ± SD (Min, Max)
Before Diwali	27 Oct to 1 Nov, 2010	39.6 ± 16.5 (2.8,65.1)	15.2 ± 8.2 (1.9,29.7)	7.1 ± 5.5 (1.2,23.5)	19.2 ± 20.1 (3.2,89.7)	755.8 ± 140 (526.9,1154.3)	990.2 ± 508.1 (526.1,2850.4)	160.8 ± 89.4 (44.2,380.4)	79 ± 39 (26.2,187.6)	1.1 ± 0.2 (0.7,1.5)	1.0 ± 0.0 (0.0,1.0)
Diwali	2 to 7 Nov, 2010	48.3 ± 13.8 (9.1,67.6)	22.8 ± 17.5 (1.7,58.9)	5.6 ± 3.9 (1.9,19.4)	19.2 ± 23 (1.9,90.8)	867.2 ± 124.3 (692.4,1219.6)	1174 ± 491.6 (744,3150.5)	211.1 ± 94.2 (48.4,413.7)	136.1 ± 95.4 (22.1,465)	1.2 ± 0.1 (1.1,6)	1.0 ± 0.0 (0.0,1.0)
After Diwali	8 to 13 Nov, 2010	19.4 ± 8.3 (1.5,33.6)	11.2 ± 10.6 (1,52.5)	6.6 ± 3.2 (2.6,17.8)	8.6 ± 6.3 (1.5,38.6)	892.4 ± 83.7 (735,1122.3)	917.1 ± 156.5 (725.5,1621.2)	157.6 ± 54.6 (52.9,311.6)	152.8 ± 95.1 (19.5,568.2)	1 ± 0.1 (0.7,1.4)	1.0 ± 0.0 (0.0,1.0)

It is worthwhile to be noticed at this stage that even then a slight reduction in the temperature of 2 to 3°C, increase in Rh values by only 5% and 27% in sunlight and night hours and nature of altering in local wind pattern of north-easterly easterly to southeasterly during pre-festival to festival, as well as continuous reduction of VC values from before Diwali to the post-Diwali episode also favor to the excessive accumulation of atmospheric pollutants near to the surface layer over Udaipur site region. In the other words, it seem too much support to the major cause of observed higher PM_{2.5} levels in the Diwali in view of interdependence of anthropogenic activities as well as with meteorological and atmospheric dynamics processes, etc., vice versa Hence, such unusual fluctuation in daily PM_{2.5} values over the festival to non-festival period may be attributed by the intensity of generation of the extra air pollutants material emission in the festival episode along with meteorological processes, vice versa.

While comparing above findings with other reported similar investigations on Diwali occasion over other Indian sites, it is an obvious fact that enhancement of air particulate pollution level of $PM_{2.5}$ by more than two times fold over Udaipur is found to be a slightly toward lower side relative to as reported elevated $PM_{2.5}$ level by 4 to 6 time excess in Diwali events relative to the control day observations at other major Indian polluted cities, i.e., Delhi, Hisar, Kolkatta, Dibrugarh, Lucknow, Agra, Varanasi etc. (Ganguly et al. (2019); Mukherjee et al. (2018); Tiwari et al. (2013); Ravindra et al. (2003); Thakur et al. (2010); Pathak et al. (2015), Barman et al. (2008 and 2009); Pachauri, et al. (2013); Pratap et al. (2021)).

3.2. Surface ozone level (O_3):

The continuous day and night time hourly O_3 variation over pre- to post-Diwali through Diwali festival periods is illustrated in Figs. 5a and 6a, respectively. The statistical parameters for day and night hours for the above period are also shown in Table 3. It is evident from figures that of noontime O_3 levels in the sunlight hours (pre-midnight hour values as shown in bracket) have shown the maximum values between 52.7 to 67.6 ppb (9.1 to 32.7 ppb) in Diwali occasion, followed by their respective lower peak magnitudes between 50.5 to 59.2 ppb (2.5 to 11.6 ppb) in before Diwali. While in post Diwali period, their corresponding lowest levels span in between 22.6 to 33.6 ppb (1.5 to 5.2 ppb). Furthermore, based on the computed statistical parameters as mentioned in Table 3, one can also easily visualize that during the Diwali period, day as well as night hours average O_3 values are also found to be enhanced by its double values, i.e., $(48.3 \pm 13.8 \text{ ppb and } 22.8 \pm 1.23 \text{ ppb})$ relative to their observed values $(19.4 \pm 8.3 \text{ ppb and } 11.2 \pm 10.6 \text{ ppb})$ on the post-Diwali period.

Thus, the impact of ignition activities on the daytime O_3 in Diwali episode is clearly observed as the enhancement in O_3 pollution by two to three-time folds in noon hours. However, it is more interesting to note at his junction that around the early night hours, their O_3 level strength also raises distinctly by five-to-six-time folds during the Diwali relative to their respective durations in the non-festival period.

To understand such observed rising of day hours O_3 level as secondary vital ambient air toxic pollutant during the Diwali episode as the festivity contribution, the main plausible primary reason is manifested as the involvement of a series of complex and mixed photochemical production reactions, such as a). $O_3 + hv$ (UV- B radiation) $\rightarrow O(1D) + O_2$; b). $O(1D) + H_2O \rightarrow 2OH$; c) $OH + CO + O_2 \rightarrow CO_2 + HO_2$; d) $HO_2 + NO \rightarrow NO_2 + OH$; e) $NO_2 + hv(\sim 420nm) \rightarrow NO + O(3P)$; f) $O_2 + O(3P) + M \rightarrow O_3 + M$; g) $O_3 + NO \rightarrow NO_2 + O_2$; h) $CO + 2O_2 \rightarrow O_3 + CO_2$; i) $CH_4 + 4O_2 \rightarrow CH_2O + H_2O + 2O_3$.

All these photochemical reaction with different chemical reaction rates involves the growing of the extra surplus prominent precursors of O_3 as an atmospheric catalyst i.e., NO_x , VCO , CO , CH_4 emissions due to biomass burning, residential and industrial activities for perturbation of primary and secondary air pollutant levels in the availability of sufficient amount of the sun's radiation intensity and presence of water vapor as an atmospheric detergent of trace gases. The similar cause of increasing day hours O_3 are also suggested by several worker (Pratap et al. (2021), more references there in).

While in night hours also, surprisingly the production of O_3 level also raised by the factor of four-time folds in the specially Diwali event relative to post-Diwali period. Such unusual enhancement in nocturnal O_3 may be well explained by the fact of the influence of intense combustion activities, e.g., surplus bursting activities of crackers, sparklers as well as biomass burning materials in artificial light decorations as the result of extra-artificial generation of light radiation from visible to infra-red region by human-made activities with surplus NO_x emissions during the night hours of Diwali celebrations, as proposed by Attri, et al. (2001). They argued that whenever firecrackers fired, it emitted a compelling proportion of light radiation from visible to infrared region, which enabled the photochemical reactions, i.e., $O_2 + O \rightarrow O_3$, photolysis of NO_2 at around 420 nm, etc. (as described by other chemical reaction in the above paragraph) for production of O_3 in nighttime Diwali activity too. It furthermore helps and witness to support nighttime O_3 formation in the absence of solar radiation instead as the result of occurring artificial light decoration and festival celebration event too (Resmi et al. (2019)).

It is obvious from the above discussions that the significant impact as a festivity contribution of extra air pollutant O_3 emissions burden in fire ignition crackers and sparklers activities, etc., have been explicitly demonstrated the plausible reasons for overall elevation in air pollution ozone emission levels as high as in day as well as also in the night hours. Such increment in O_3 is quite well matched with the coincidence of the timing with peak fire celebration activity interval, i.e., from evening to pre-midnight hours in Diwali relative to similar interval of post and pre-Diwali period.

While comparing to currently reported finding with other observed result at other sites, the overall increment of ground ozone value in Diwali festival relative to control days has also been reported by Pathak et al., 2013 in Dibrugarh. They reported that O_3 level raised up to 43 ppb, which is a slightly lower than relative to the observed higher ozone value of 67.5 ppb at Udaipur. In contrast to the above finding, Yerramsetty et al. (2013), Ganguly. (2009), Attri et al. (2001), Saha et al.(2014), Pachauri et al. (2013) and Ganguly et al.(2019) have found an increase in day hours ozone concentration values go beyond above the prescribed limit of NAAQS value of 120 ppb over prime urban polluted Indian cities in Diwali festival. But, at Udaipur and Dibrugarh, Varanasi city, such elevation ozone level during the Diwali episode could not touch to go beyond the NAAQS prescribed healthy and safest limit O_3 value (Pratap et al. (2021)).

3.3. Carbon monoxide (CO):

The main potential emissions source of CO is recognized as biomass burning gas materials (Lal et al. (2008)), which is likely to be enhanced in injecting the exhausts of burning materials on the fireworks, sparklers, as well as artificial and natural light decoration activities on occasion of the Diwali festival. From their day time temporal behavior as depicted in Fig. 5b, the minimum CO levels in the noon to afternoon hours (maximum CO magnitudes in morning hours as shown in bracket) exhibit a growing trend of 526.9 ppb (1154.6 ppb) during before Diwali to their remarkable rising level of 719 ppb (1219 ppb) in main Diwali days celebration. But after Diwali days, their respective concentrations show a declining trend of 715 ppb (1127 ppb), which may be expected with simultaneously reduction of extra anthropogenic pollution activities from the Diwali festival period to after the festival period.

From a close look at the analysis of maximum as well as minimum night hours CO levels in festival and non-festival period as depicted in Fig. 6b, it may be noticed that the primary peak of CO values in early night hours (dip level in early sunrise hours as mentioned in bracket) showed the increasing feature of

2856 ppb (1526 ppb) in pre-Diwali and to reach their peak levels of 3150 ppb (744 ppb) in the main Diwali event. Later, their values were found to be declined to 1621 ppb (725 ppb) on post-Diwali. This feature may be expected to follow in view of the reduction of anthropogenic emission pollution strength during the pre-Diwali festival period and post-festival period relative to the peak festival period.

In addition to this, it is clearly evident from Table 3 that average day hours CO values are found to be the highest mean 867.2 ± 124.3 ppb in the Diwali festival in reference to that their respective lower mean values 755.8 ± 140 ppb and 763 ± 201.2 ppb, in pre as well as in post-Diwali day, respectively. In the same way, in the night hour period, the CO magnitudes also exhibited the maximum mean value 1174 ± 491.6 ppb in Diwali, followed by their intermediate mean value of 990.2 ± 508.1 ppb on pre-festival and their lower mean value 917.1 ± 156.5 ppb during the post-festival period.

From the ongoing discussions, it is quite imperative to say that relative changes in the night time maximum CO concentration levels for Diwali to after Diwali seem to be more remarkable enhancement, which is fluctuated up to 100% higher relative to their corresponding insignificant changes in the maximum day hours CO levels only of about 20%. Such typical rising level in the mean CO is higher about 50% during night hours during the Diwali celebration interval, whenever compared to before and after the Diwali episode period. It gave furthermore ample support of enunciated influence of additional biomass burning emissions activities by an additional local man-made activity around the observing site during the night hour festival period.

While discussing the present CO observations with other earlier already reported CO measurements at Dibrugarh by Pathak et al. (2013), Pratap et al. (2021) at Varanasi and Ganguly et al. (2019) over various Indian cities in Diwali festival period, it is observed a that the night time peak CO value at Udaipur was noticed to be quite in a higher range from 1621.2 ppb to 3150 ppb (~ 100%) during Diwali to post festival days then that the reported respective maximum CO value range from 1180 ppb to 1420 ppb (19%) during normal day to Diwali festival days over Dibrugarh (Pathak et al. (2013)). It is an interesting fact that the daily mean hourly CO values at Udaipur always remain to be lower than that of the prescribed limit NAAQS CO value of 3400 ppb and its two-time higher value at Varanasi in the festival period (Pratap et al. (2021)).

3.4. Oxides of Nitrogen (NO_x):

The NO_x is a key ambient air pollutant, which is known to be main chemically reactive as a photochemical-oxidant, a health-hazardous gas as well as one of the precursors of O₃. The sources of atmospheric NO_x vary from rural to city's locations. As NO_x is generated from various sources viz., from the exhaust of transport, fossil fuel ignition, thermal power plants, industry emission activities to the atmosphere as NO, which is converted into NO₂ and other nitrogen species leading the formation of various photochemical oxidants (O₃, N₂O, PAN) with OH⁺ within the troposphere (Gupta et al. (2003)). Thus, NO_x, CO, and BC are accepted as the main byproducts key anthropogenic ambient pollutants of burning events as well as prominent precursors for the photochemical catalyst of O₃ generation in the troposphere, which has been identified as tracers for anthropogenic emissions activities strength too.

Therefore, firework display events can become more sensitive to display the immediate response in the short-term fluctuations in primary and secondary atmospheric pollutants NO_x, CO, BC, and O₃ loadings on Diwali eve. So, the day and night time temporal variations in NO_x level are illustrated for pre-Diwali to post-Diwali through Diwali events day periods in Fig. 7 (a) and 8 (a), respectively.

Figure 7(a) shows that maximum day hour NO_x concentrations, in after just sunrise hours, vary in higher ranges from 15.5 to 23.5ppb and 11.4 to 17.8 ppb during before and after the festival occasion relative to as observed its lower range from 9.8 to 14.5 ppb on Diwali days, In contrast to this, daytime minimum values around noon hours remain almost the same and lie only in the ranges from 1.2 to 2.7 ppb, 1.9 to 2.8 ppb and 2.1 to 3.6 ppb during before, Diwali and after festival days, respectively.

As far as concerned to the night hours temporal evolutions, it is seen from Fig. 8b that just after the sunset hours, the ambient atmospheric NO_x mixing ratio began to rise very abruptly with their varying magnitudes and attained the different distinct primary peak values from 40 to 85 ppb and 16 to 96 ppb from 19:30 to 22:00 hrs, i.e., coinciding with the timing of the main celebration firework display activity duration in pre-Diwali to the main festival period. Meanwhile, after the Diwali occasion, the such peak NO_x values were recorded in the lowest range from 13 to 35.6 ppb. At the same interval, the minimum nocturnal NO_x values did not show much change and were recorded within a small range between 2.6 to 5.2 ppb only over the festival and non-festival period.

From the above discussion, it is obvious that the night time peak value of NO_x level displays an abrupt increase in behavior from their maximum levels as high as typical value of 96 ppb only on firecracker burning activity interval on before and the main festival period relative to as high as the only 38.6 ppb after the Diwali night period. Accordingly, this significant amount of nighttime enhancement in NO_x concentration about 50 ppb, i.e., more than the two times value on Diwali festival comparative to after the festival period gives ample support and direct confirmation of the impact of huge amounts of surplus accumulation of NO_x emissions owing to fossil fuel and bio-mass burning as the result of the ignition and bursting of firecrackers episodes during the nighttime Diwali celebration interval.

Additionally, the peak values of nighttime NO_x as result of extra generation of NO_x due to festivity celebration to be appeared to increase by double values in Diwali. Whereas in the case of daytime and in contrary to night time NO_x behavior, the peak value of day time NO_x reduces by 6 to 9 ppb, mainly attributing by the fact of the more consumption of NO_x by well-known chemical titration reaction for production of O₃ in day hours in more abundance of solar radiation in day hours as discussed in earlier section too. In other words, it also leads to more consumption in terms of O₃ production by photo chemical reaction before by photolysis chemical titration mechanisms of conversion of NO₂ → NO + O under solar UV to IR radiation intensity in the sunlight hours with certain delay in completing several series of photochemical reactions as discussed and written in earlier section.

It is to be noted here that the typical temporal NO_x variation, which is in the opposite nature of the diurnal ozone behavior and like the CO diurnal curve shows the well-defined and prominent visible effect in special reference to night hours relative to the day hours as the induced firework effect activities of Diwali festival. Based on the analysis of derived statistical parameters, it also suggests similar fact that night time average values with a standard deviation of NO_x

are observed to be higher during Diwali and before Diwali eve, when compared to after the festival period. While in the case of the day hours, the average and standard deviation values do not show any significant change from before Diwali to after Diwali through main festival days.

While comparing to present study with other similar investigations reported for firework induced change in NO_x and NO₂ emission level during Diwali festival over other Indian stations, i.e., Delhi, Dibrugarh, Kolkatta, Lucknow, Hyderabad, Agra, Varanasi and other Indian sites (Singh et al. (2010); Ganguly. (2009); Pathak et al. (2013); Saha et al. (2014); Yerramsetty et al. (2013); Pachauri et al. (2013) and Ganguly et al. (2019); Pratap et al. 2021)), it is observed that the diurnal NO_x nature is seen to be consistently identical in shape with their observed diurnal NO_x behavior in festival to the non-festival period over other Indian sites. But such increasing NO_x levels are quite high, i.e., two to six times higher over polluted and densely populated urban locations relative to as observed the marginal enhancement by a two-time fold of NO_x level at Udaipur. However, the maximum level the NO_x level is seen in the above prescribed safest and health limit of NAAQS 24 hourly values of 45 ppb over other major Indian urban centers only, but not observed in Udaipur city (Singh et al. (2010); Ganguly, (2009); Pathak et al., (2013); Saha et al, (2014); Yerramsetty et al. (2013); Pratap et al. (2021)).

3.5. Ratio of CO/NO_x:

Lal et al. (2008) has shown that CO, NO_x, VOC, Hydrocarbons etc., are well known major prominent precursors for the photochemical O₃ production as well as for anthropogenic emissions tracer, as can be seen from various photochemical reactions in the last section. They have also reported that the primary emission CO source is mainly linked with low-temperature incomplete combustion processes of the bio-mass burning emission activities (e.g., fireworks, artificial light decoration activities, forest fires, vehicular exhausts, gas cooking stove, industrial emission activities, etc.) and the NO_x emission sources with the higher temperature incomplete combustion activities (e.g., use of fossil fuel in vehicles, transportation, power plants, burning of crackers, sparklers, artificial light decoration events, etc.). So, the CO/NO_x ratio parameter serves as a possible indicator of evaluating their relative role of bio-mass burning emission sources concerning to fossil fuel emission source processes as well as in terms of mobile and fixed source emission activities of their two compound sources. Based on earlier reported studies, typical $\Delta\text{CO}/\Delta\text{NO}_x$ values were observed of 30–40 ppb/ppb near some rural sites in China (Wang et al., 2007) and their higher ratio values over Delhi (Aneja et al., 2001). The higher CO/NO_x ratio values were also reported the more predominant nature of mobile source strengths of these pollutants over the urban region like as in Delhi by Aneja et al. (2001).

Thus, in high-temperature of inefficient combustion processes, these ratio values are registered to be lowering than those values in case of emitted from biomass burning. However, in the case of low-temperature incomplete combustion, its values are on the higher side. In other words, the ratio of CO and NO_x is mainly controlled by the dominance of the origin and nature of burning material sources, i.e., biomass and fossil fuel substances, etc.

In view of identifying the relative percentage share of the influence of firework emission by bio-masses concerning to fossil fuels origin activities during the Diwali festival as well as the non-festival occasion, the day and night hours of CO/NO_x values are plotted with function to time in Figs. 7(b) and 8(b).

From the first glance to figure, it has appeared from Fig. 7(b) that the minimum early morning hours CO/NO_x values are seen to be altered in the small ranges from 44.2 to 59 and 53.1 to 70.8 prior and after the festival occasion, respectively, While in the case of prime Diwali event, the respective values spread in slightly higher ranges from 65.6 to 88, indicating dominance of mobile sources of their compound. As far as concerned, their maximum CO/NO_x values in late noon hours spread from 271 to 380 and 228 to 311 during before and after Diwali days to as peak value as in the range of 322 to 416 during the Diwali festival. Thus, day hour feature demonstrates the slight enhancement contribution in bio-mass burning emissions in the festival episode relative to fossil fuel ignition emission activities in non-festival period.

While in the case of nocturnal hours, the nocturnal maximum of CO/NO_x values in pre-mid night hours are between 161 to 568.2 during the Light festival celebration period and their observed intermediate values of 223.4 to 255.3 after Diwali. These values decrease towards a lower range from 134 to 177.4 in before Diwali period, as shown in Fig. 8b. At the same time, the minimum night-time values around post-midnight hours did not show any large marked variability from pre-to post-Diwali, whereas in the case of Diwali occasion, the corresponding range occurred in between 23 to 106.8, which was slightly above the observed minimum nighttime range relative to non-festival period.

Such observed maximum values of CO by NO_x in night as well as in daytime above exceeding the 150 has given clear confirmation of the prominent fire-works induced effects of the Diwali episode in term of more enhancement of bio-mass burning processes relative to fossil-fuel activities along with their pre-dominant mobile source activities.

Hence, the night time CO/NO_x evolution is noticed to be altered mainly in accordance as per the relative strength of direct bio-mass material activities from sunset to pre midnight hours. However, in daytime conditions, their values may be mixed contributed by (i) surplus burning activities in Diwali event, (ii) low efficiency of converting the CO due to their less consumption into other chemical species into O₃, etc., or more residence lifetime of CO relative to NO_x, (iii) the high efficiency of photochemical breaking up of NO_x in O₃ generation, etc., or combined effect of all the above possible factors of controlling the day time fluctuation in CO/NO_x.

3.6. Angstrom Exponent Coefficient of BC absorption coefficient (α_{abs}):

Here, α_{abs} is considered one of the other important leading parameters of giving the information about the origin of anthropogenic emission materials segregated based on their magnitudes (Schnaiter et al. (2003); Kirchstetter et al. (2004); Gadhavi and Jayaraman. (2010); Clarke et al. (2003)).

It is seen from Fig. 9a that the maximum morning hour's α_{abs} magnitudes are appeared to be in the highest range of 1.37 to 1.55 in Diwali occasion relative to their observed lower range of 1.1 to 1.4 during non-festival period. Whereas, the minimum α_{abs} values around noon hours persists in a lower range of 0.99 to 1.06 on Diwali celebration as compared to their lowest range of 0.7 to 1.04 over the non-festival as shown in Fig. 9b. Such observed higher day α_{abs} values

during the festival days relative to non-festival eve has given a direct indication of surplus anthropogenic emission activities of extra bio-mass burning materials relative to incomplete combustion fossil fuel ignition materials.

However, as expected, in the nighttime, the α_{abs} values raised from sunset to around midnight hours, after that, their magnitude declined till early morning hours, because of the possible argument of excessive emission of exhausts of bio-mass burning materials in light decoration and other activities during the Diwali celebration. The minimum α_{abs} levels in after midnight hours occur in a lower range between 0.82 to 1.1 during non-festival as well as festival night time, while in early nighttime festival duration, their maximum values spread in the highest range from 1.4 to 1.55, but in after and before the event, their values are in a larger range from 1.1 to 1.35. It is very interesting to note that such similar observed α_{abs} values in ranging from 1.2 to 1.5 was also reported by Goel et al. (2021) from the BC aerosol measurements at Delhi during bio-mass burning activity in Diwali episode.

Such elevated night time α_{abs} values along with their higher magnitudes above 1.45 on prime Diwali occasion explicitly witness the influence of origin about more air pollutants aerosols generation emissions level leading to firing crackers and sparklers as anthropogenic emission activities, i.e., from biomass burning events, whereas, the lower α_{abs} values below to 1.1 in after and before Diwali even manifest the effects of fossil fuel emission concerning of incomplete combustion of vehicular motor emission activity, etc.,

Thus, the major predominant role of emission of anthropogenic air pollutant sources in the festival period is biomass emission activities, such as the Diwali day as compared to fossil fuel activity, which seems to be more prevalent in the preceding and subsequent days of Diwali. It can say that percentage of vehicular emission activity shows a lesser impact than that of ignition of fireworks activity specifically on Diwali days.

3.7. Role Of The Regional Scale Of Air Pollution Emission Source Activities:

Despite their well identifications of the primary role about local emissions sources on the air pollutants strengths variability during the anthropogenic episode, the regional air pollutants emission activities have been also familiar as one of another leading factor of influencing the seasonal atmospheric pollutants characteristics even than over less environmental polluted and low-density populated sites, because of atmospheric pollutant outflows and long-distance transports of air pollutants from remotely heavily polluted source regions to less polluted and clean location, known as intercontinental air pollutants transportation phenomena (Moorthy et al. (2003); Gogoi et al. (2008); Nair et al. (2007); Beig et al. (2007); Vinoj et al. (2010); Lawrence & Lelieveld. (2010)).

In this context, computed 72 hourly interval backward wind trajectories for the period from pre-Diwali, Diwali and post-Diwali period are accessed for observing site at low advection level of 500m ASL (a well within the atmospheric boundary layer) and another at a high advection level of 4000m ASL (in the lower free tropospheric region) from the NOAA HYSPLIT (Hybrid Single-Particle Lagrangian Integrated Trajectory) model4 (http://www.arl.noaa.gov/HYSPLIT_info.php; Draxler & Roliph. (2003)).

It is obvious from Figs. 10(a-c) that, in the case of low-level advection altitude of 500m of backward air mass trajectories, their source regions of travel air mass pathways are concerned to coming from to heavily populated and industrialized regions with the dominated human-made activity of IGP region. Hence, the parcel of the above backward atmospheric wind can carry atmospheric pollutants from remotely continental location of the heavily polluted source region in North and Central India during normal as well as in the episode period. But, in the case of travelled path regions of air mass trajectories of higher altitudes of 4000m in Figs. 10(d-f), especially in post-Diwali cases, dispersion of air mass movement brings to suppose lower atmospheric pollutants along associated with air masses flowing primarily over marine environmental or clean humid regions of the Indian Ocean and Arabian Ocean, which are predominantly concern with the category of clean and humid environmental conditions. At the same time in case of pre-Diwali to Diwali, such air mass passed through the mainland region of concern highly human-made polluted and industrial activity zones.

Hence, the backward air mass trajectories of higher altitude level concerning of longer path areas, primarily circulated over the marine and oceanic sea source region, whereas in case of lower altitude of the advection pathway region of shorter duration of backward air mass flow pattern travel entirely over the mainland and continental air polluted regime during the completion of festival event. As the results of the above fact, it is expected that during after Diwali period, sharp reduction in air pollutants level after Diwali may be attributed to long range transportation of lesser and clean air pollutants along with higher humidity, i.e., above Rh values of 70% and strong wind from the oceanic region relative to excess air pollutant loading in pre-Diwali and Diwali festival period, which may give a clear support demonstration of regional emission source activity effect as observed in this study.

4. Conclusions

From the assessment over all change in various atmospheric air pollutants level and their types of air pollutant emission sources in artificial light display activities, namely, fireworks, crackers, etc., during the Diwali celebration, the following main results are summarized as below:

- a. The clear and distinct sharp increment by 1.5 to 3 times in O_3 , NO_x , CO levels as air pollutant strength level along with several source types and discrimination key parameters are noticed during main festival celebration period from evening to pre-midnight hours of the Diwali period relative to their observed respective levels on non-festival period. It clearly demonstrates the main possible reason of altering the air pollutant source and strength variability, which may be attributed only from extra emissions of precursor gases, air pollution level in the concurrence of peak interval with fireworks displays celebration event concerning primarily by the biomass activity than that observed fossil fuel emission source activity.
- b. One of most prominent influenced air pollutant parameters i.e., ambient suspended particulate matters $PM_{2.5}$ is found to be increased above the double value of NAAQS $PM_{2.5}$ level during the Diwali festival period. Further, it indicates a serious impact in atmospheric pollution levels by enhancement of particulate matter levels above prescribed healthy safest level during the Diwali occasion even then over low populated density cities like at Udaipur.

- c. The strength of day hour ambient air pollutant O_3 , values only is seen to be enhanced marginally from 70 to 120% and α_{abs} and CO/NO_x magnitudes from 0.9 to 1.5 and 104 to 560, during the Diwali period over their normal day period. It supports clear evidence of the more predominance influence about burning emission sources due to mainly by bio-mass activity in festival occasion and more fossil fuel emission activity in non-festival days.
- d. The night hour maximum NO_x, CO, α_{abs} and CO/NO_x values are raised by greater than double values relative from their corresponding control days level in the presence of visible radiation generated by display of burning of firecrackers, sparklers, artificial lightning during the Diwali festivity period relative to similar period of the other non- festival period. As the possible result of this, an increase in the nighttime production of O_3 by 90% is also seen due to photochemical reactions in Diwali night fire decoration and celebration period.
- e. The sharp reduction in air pollution level during the post-Diwali could be well explained in view of alteration in different air mass pathway duration and their different type source regions specifically in case of higher altitude level 4000 m. The higher altitude advection pathways wind trajectories may bring the lesser air pollutants along with highly moisture nature wind flow from oceanic region, While in pre-Diwali to Diwali period, their observations can logically support to carry out the large amount of extra air pollutants burden from the different anthropogenic surrounding and adjacent source regions before reaching to a measurement site.

Declarations

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Data availability Data and material are available upon request from BRIJ MOHAN VYAS (bmvyas@yahoo.com).

Compliance with ethical standards

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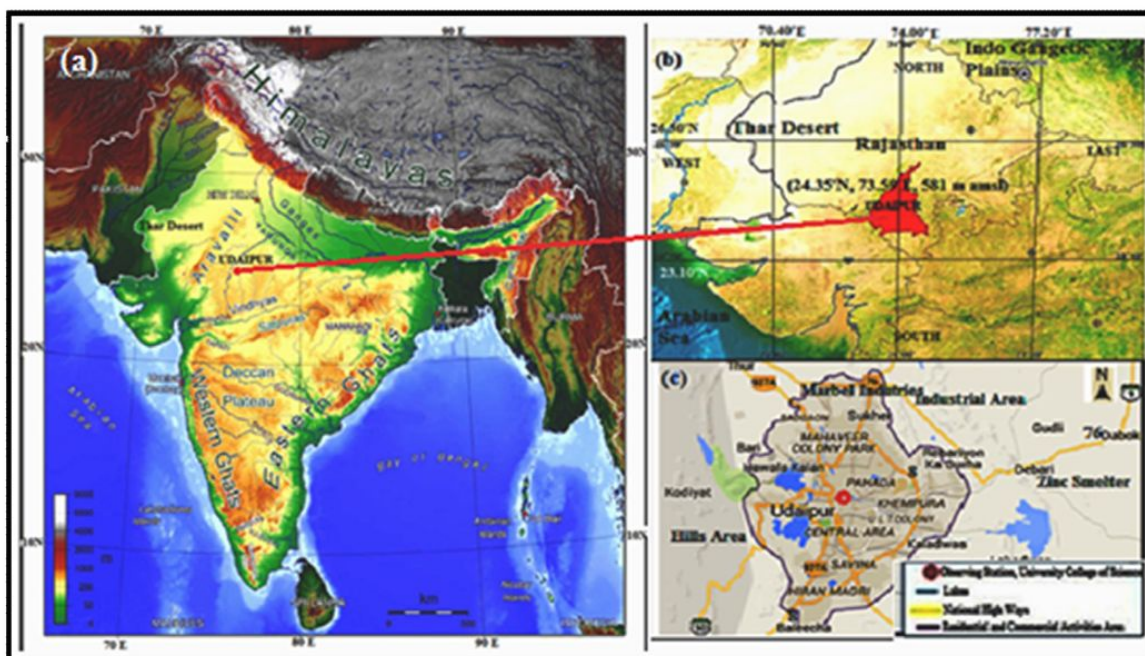
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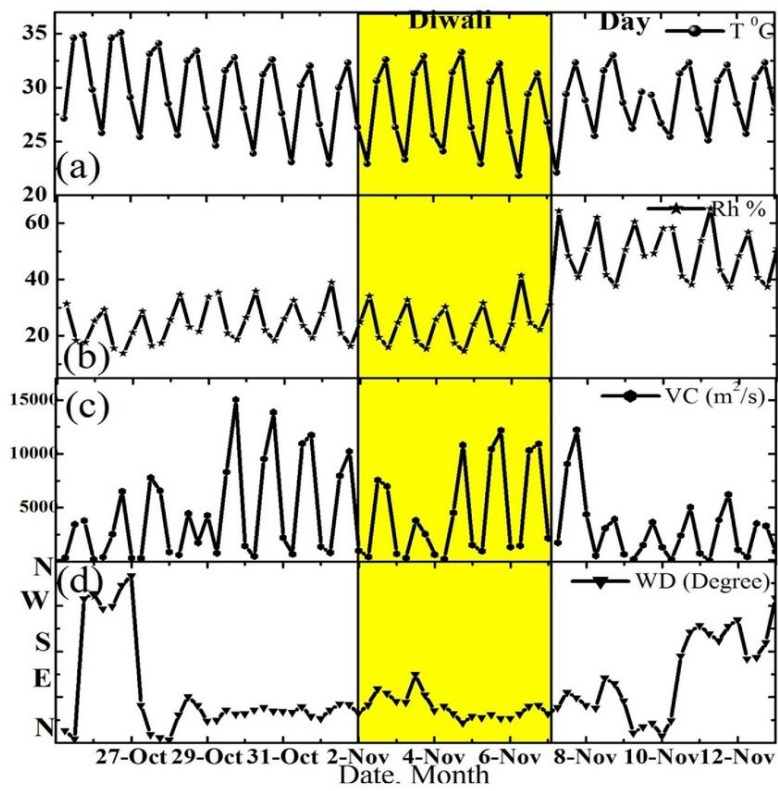
Figures



Figs. 1 (a-c):

Figure 1

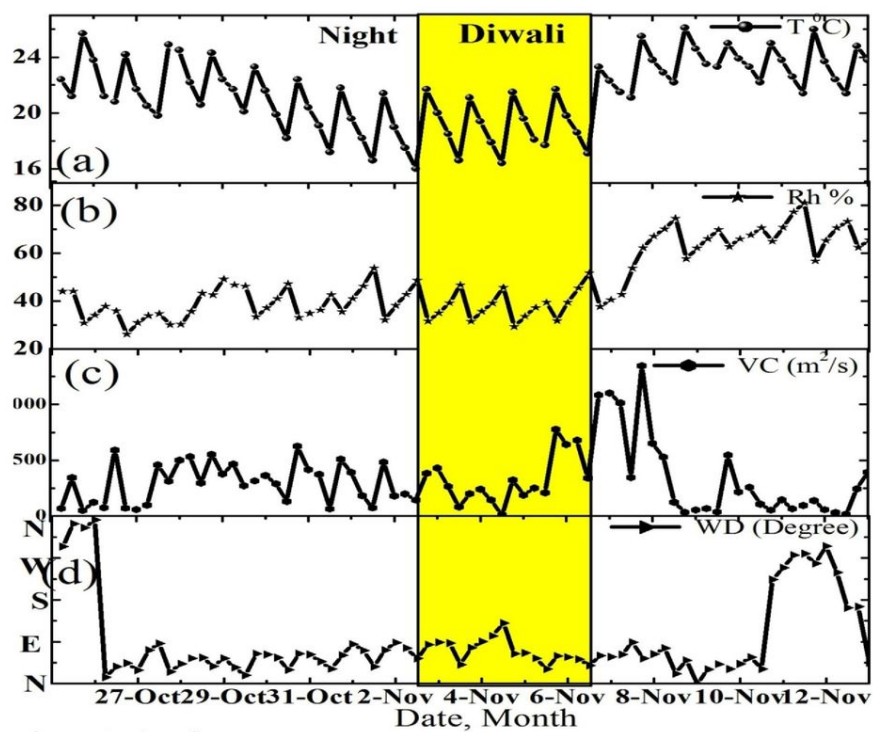
(a -c) : (a) Geographical Indian map indicating the experimental location and (b & c) physical map around Udaipur and local site map showing the details around observing site.



Figs. 2(a-d):

Figure 2

(a-d): Temporal sunlight hourly variations in T , Rh , Vc and WD measured on before, main and after Diwali festival days.



Figs.3 (a-d):

Figure 3

(a-d): Temporal nocturnal hourly variations in T , R_h , V_c and W_D measured on before, main and after Diwali festival days.

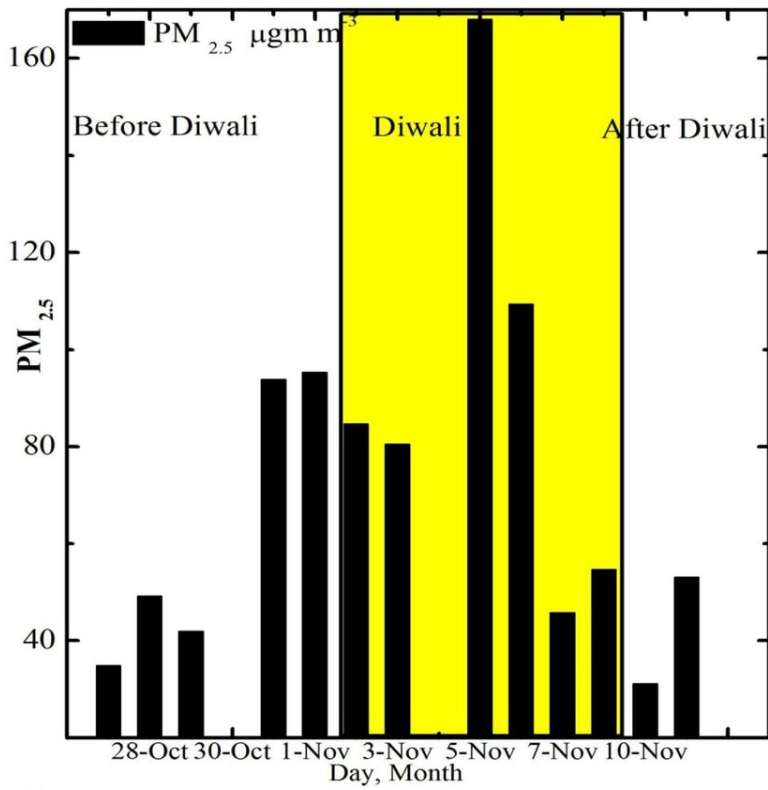
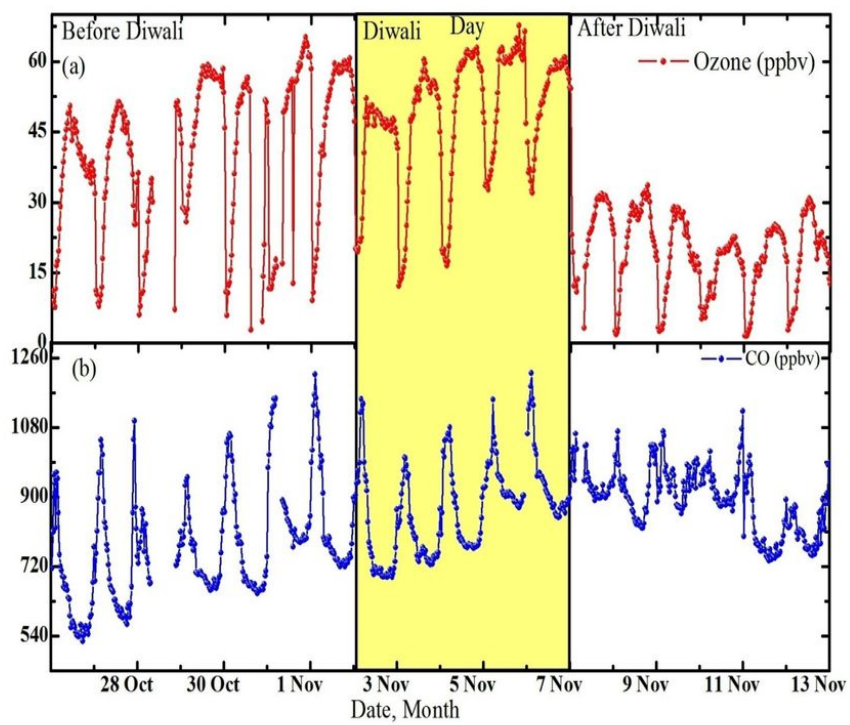


Fig. 4:

Figure 4

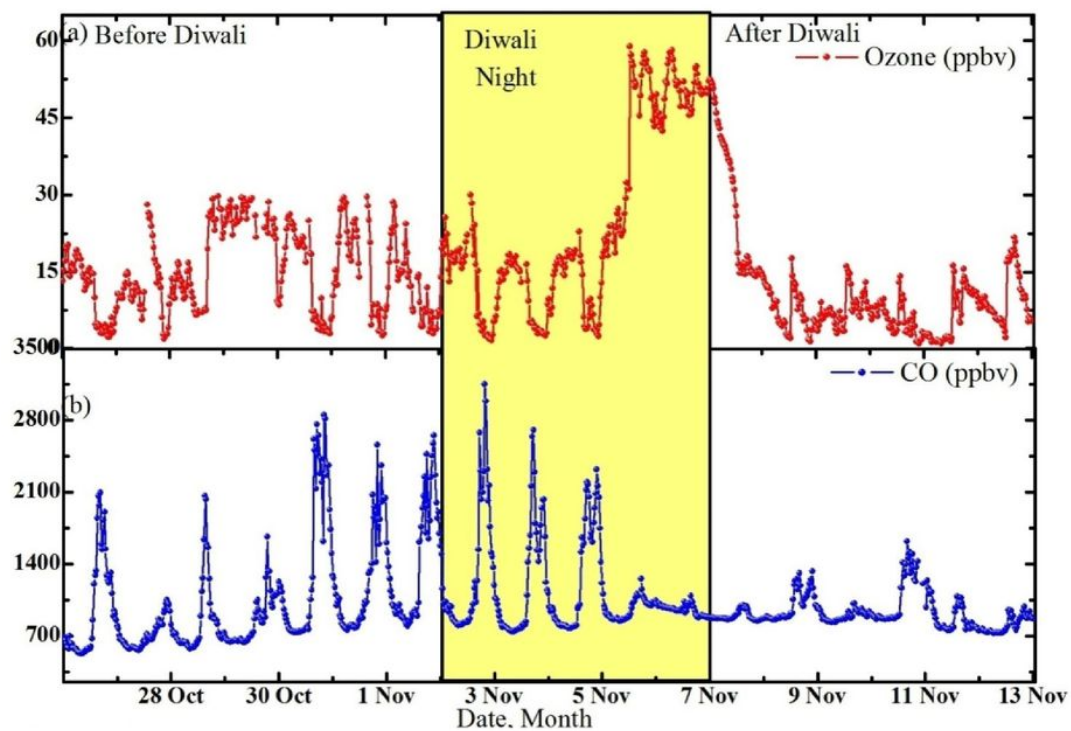
Day to day changes in 24 hourly average $PM_{2.5}$ levels from pre-Diwali to post-Diwali through main festival days.



Figs. 5(a-b):

Figure 5

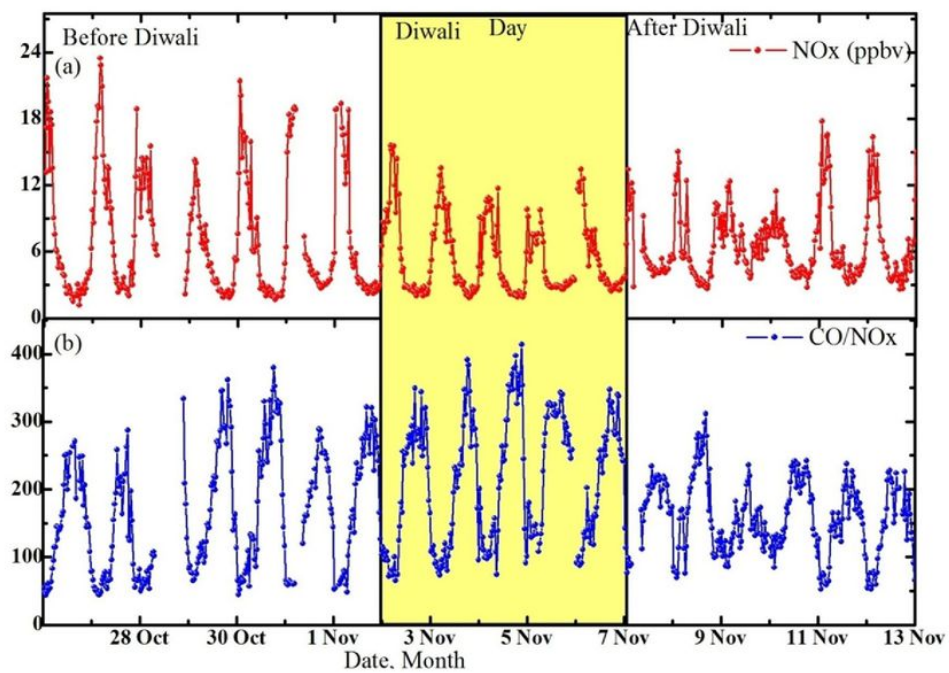
(a & b): Temporal daytime hourly variations of surface ozone and carbon monoxide measured on before Diwali, Diwali and after Diwali festival dates.



Figs.6 (a-b):

Figure 6

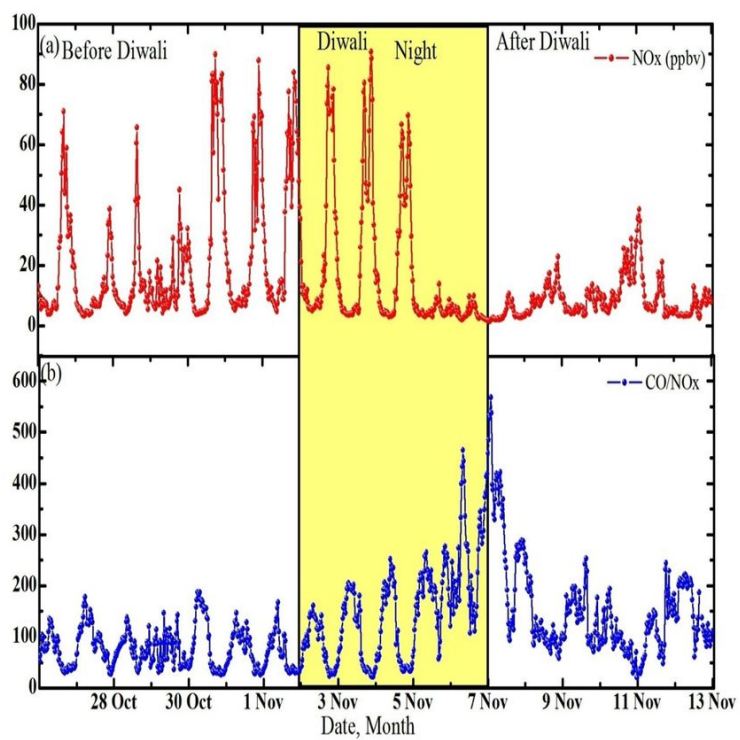
(a & b): Temporal nighttime hourly variation of surface ozone and carbon monoxide measured on before Diwali, Diwali and after Diwali festival dates.



Figs.7 (a-b):

Figure 7

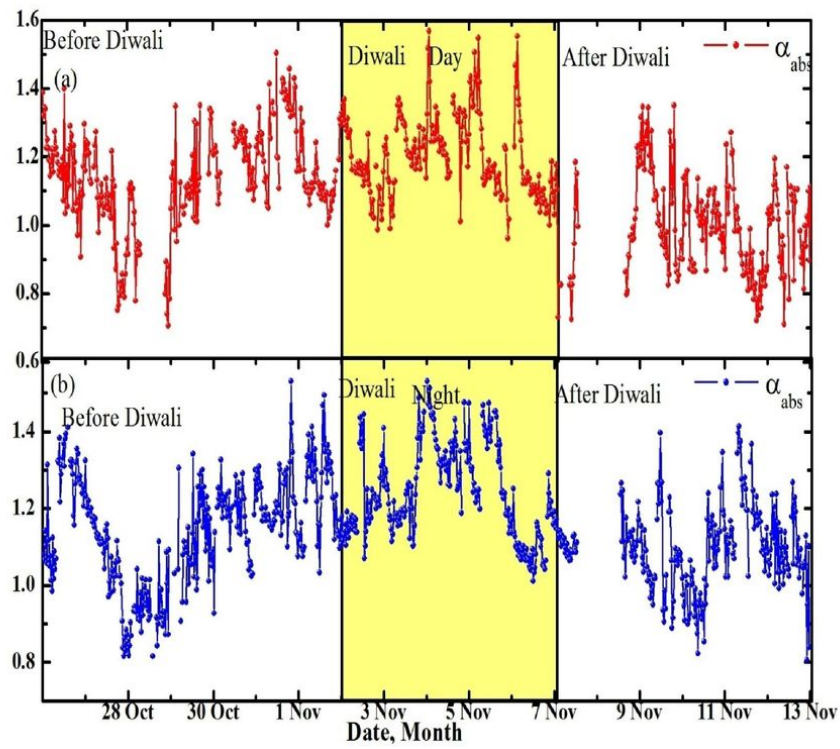
(a-b): Temporal daytime hourly variations in NO_x and CO/NO_x levels on before Diwali, Diwali and after Diwali festival dates.



Figs. 8 (a-b):

Figure 8

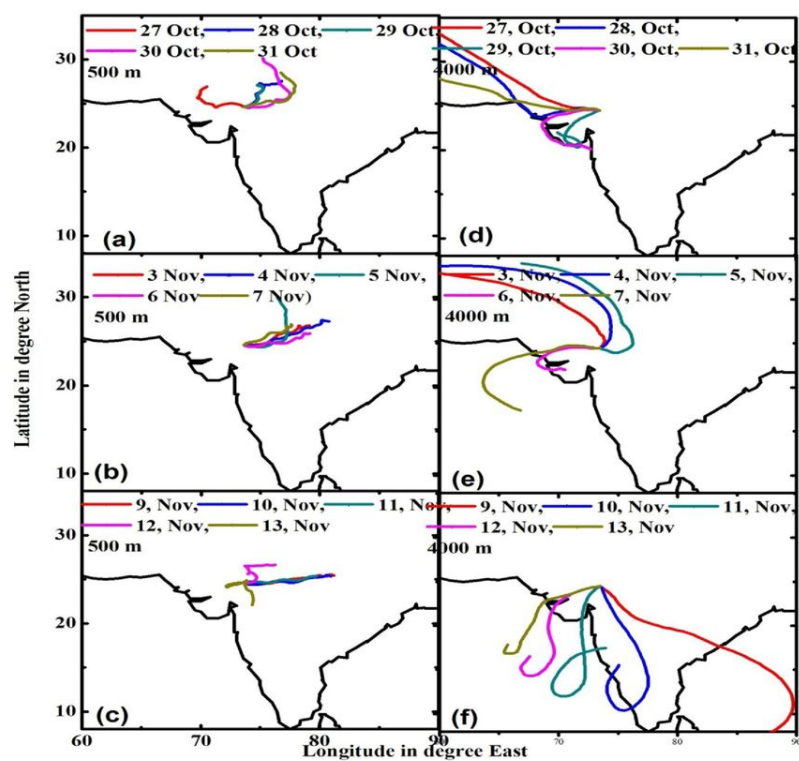
(a-b): Temporal nighttime hourly variations in NO_x and CO/NO_x values on before Diwali, Diwali and after Diwali festival dates.



Figs. 9 (a-b):

Figure 9

(a-b): Temporal day and nighttime hourly variations in α_{abs} calculated on before Diwali, Diwali and after Diwali festival dates.



Figs. 10 (a-f):

Figure 10

(a-f): Air Mass Backward Trajectories using NOAA Hysplit Model for 500m AGL and 4000m AGL observed over Udaipur on pre-Diwali, post Diwali as well as on main Diwali dates.